

EECE 571Q

Lecture 15: Decoding

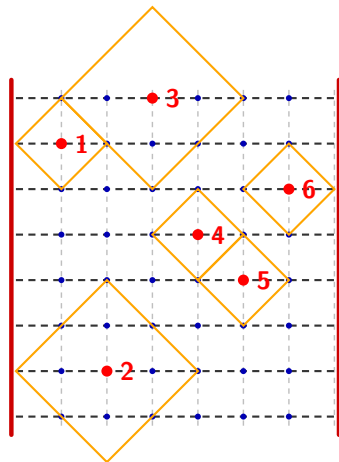
Thursday 16 July 2026

Quiz: Solution to Question 1

- ① An experiment requires a quantum decoder with strong logical error rate suppression. Runtime matters but is not as important as accuracy and scalability since we are not running in real time. Which decoder is most appropriate? **BP+OSD**
- ② A quantum device requires fast decoding that runs in real time on a trapped ion system. Accuracy is important but it needs to avoid a large growth in decoding time. Which decoder is most appropriate? **BP+LSD**

Quiz: Solution to Question 2

■ Union-find algorithm after termination



Lecture Outline

- 1 Quiz Solutions
- 2 Introduction
- 3 Selected Neural Network Architectures
 - Convolutional Neural Networks
 - Recurrent Neural Networks and Transformers
 - Graph Neural Networks
- 4 Practical Implementations
 - GNN Example
 - AlphaQubit 1
 - AlphaQubit 2

Learning Outcomes

- **L06:** Describe opportunities for machine learning to enter the quantum decoding pipeline
- **L07:** Distinguish between ML decoder architectures such as CNNs, GNNs, and transformers
- **L08:** Justify which ML decoder architecture should be chosen given the benefits and limitations of each method and the QEC code

Last time

- Tanner Graphs
- Minimum Weight Perfect Matching (MWPM) and the Blossom algorithm
- Detector error model (DEM) and performance metrics
- Belief Propagation BP and variants (OSD, LSD)

Maximum likelihood decoding (review)

Probability of logical class given syndrome sequence $\mathbf{s}$:

$$\Pr(\ell \mid \mathbf{s}) = \frac{1}{\Pr(\mathbf{s})} \sum_{E \in C_\ell(\mathbf{s})} \Pr(E) \propto Z_\ell(\mathbf{s})$$

Choosing error with the highest probability of occurring:

$$\hat{\ell}_{\text{MLD}} = \arg \max_{\ell \in \{0,1\}^{2k}} Z_\ell(\mathbf{s}) \quad Z_\ell(\mathbf{s}) := \sum_{E \in C_\ell(\mathbf{s})} \Pr(E),$$

Requirements for real time decoding (review)

If $r_{gen}/r_{proc} = f > 1$, a small initial backlog in processing syndrome data will lead to an exponential slow down during the computation. [Terhal 2015]

Limits of Classical Algorithms

- MWPM does not consider the whole *coset* probability
- Usually developed under simplified noise models
- Not accounting for leakage

Idea: Let the data speak for itself!

- Train a complex model, with minimal structural hints to learn from data
- Tradeoff between high training cost and better accuracy

What can we do? (review)

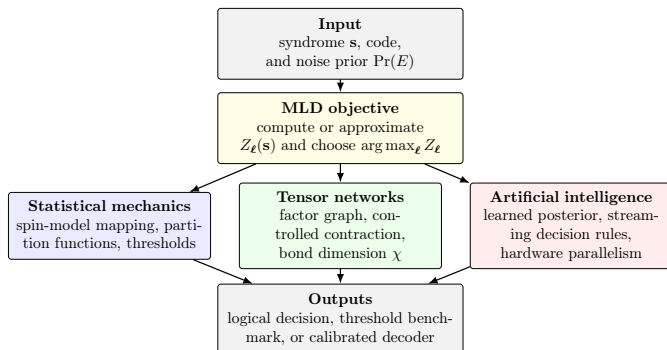


Figure: A classification of the recent approaches to quantum decoding (Fig. 2)¹

¹Hanyan Cao et al. (2026). *Maximum Likelihood Decoding of Quantum Error Correction Codes*. [arXiv: 2605.17230](https://arxiv.org/abs/2605.17230).

Convolutional Neural Network

Is a fully-connected network not sufficient?

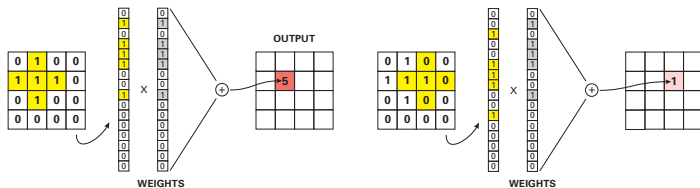


Figure: Fully connected network on an image (Fig. 14.1²)

²Kevin P. Murphy (2022). *Probabilistic Machine Learning: An introduction*.

Convolutional Neural Network (continued)

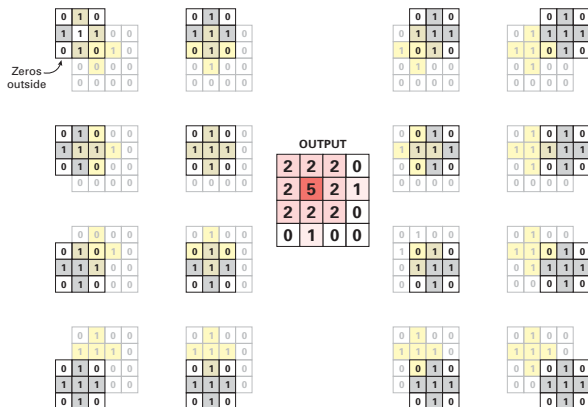


Figure: Cross-correlation on an image (Fig. 14.7)³

³Kevin P. Murphy (2022). *Probabilistic Machine Learning: An introduction*.

Convolutional Layer Variants

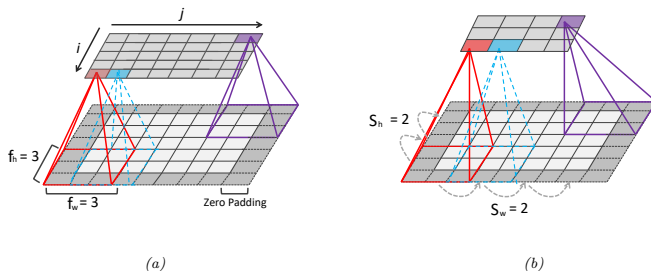


Figure: a) Convolution with zero padding. b) Strided Convolution (Fig. 14.8)⁴

⁴Kevin P. Murphy (2022). *Probabilistic Machine Learning: An introduction*.

Convolutional Layer Variants (continued)

- Increases receptive field
- Leaves compute unchanged

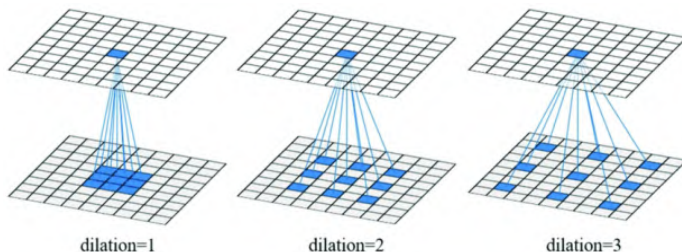


Figure: Dilated convolution for 3×3 filter⁵

⁵Ximin Cui et al. (2019). "Multiscale Spatial-Spectral Convolutional Network with Image-Based Framework for Hyperspectral Imagery Classification". In: *Remote Sensing* 11.19. DOI: [10.3390/rs11192220](https://doi.org/10.3390/rs11192220).

CNN Pipeline Example

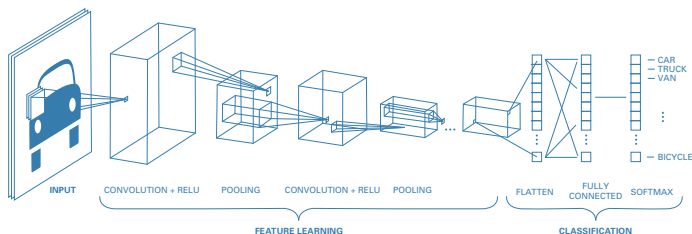
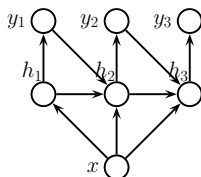


Figure: Example of a CNN for image classification (Fig. 14.13)⁶

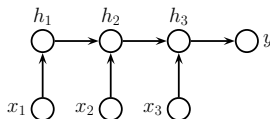
⁶Kevin P. Murphy (2022). *Probabilistic Machine Learning: An introduction*.

Recurrent Neural Network (RNN)

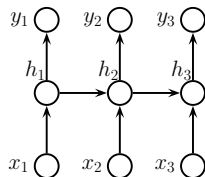
- To process sequential data such as timeseries and text
- Shares network parameters across time



(a) Vector to Sequence



(b) Sequence to Vector



(c) Sequence to Sequence

Figure: Types of RNN encoding and decoding (Chapter 15)⁷

⁷Kevin P. Murphy (2022). *Probabilistic Machine Learning: An introduction*.

Attention and Transformers

Attention can be looked at as a soft dictionary lookup:

$$\text{Attn}(\mathbf{q}, (\mathbf{k}_1, \mathbf{v}_1), \dots, (\mathbf{k}_m, \mathbf{v}_m)) = \sum_{i=1}^m \alpha_i(\mathbf{q}, \mathbf{k}_{1:m}) \mathbf{v}_i \in \mathbb{R}^v,$$

where $0 \leq \alpha_i(\mathbf{q}, \mathbf{k}_{1:m}) \leq 1$ for each i and $\sum_i \alpha_i(\mathbf{q}, \mathbf{k}_{1:m}) = 1$.

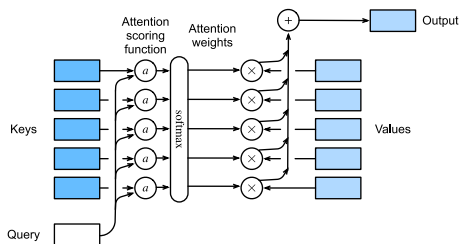


Figure: Visual of how attention works (Fig. 15.16)⁸

⁸Kevin P. Murphy (2022). *Probabilistic Machine Learning: An introduction*.

Attention and Transformers (continued)

- Often, α_i are chosen from a softmax distribution.

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathcal{S} \left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}} \right) \mathbf{V}$$

- Self attention: Variables $\mathbf{K}, \mathbf{Q}, \mathbf{V}$ depend on input only
- Cross-attention: Keys, values depend on a second sequence

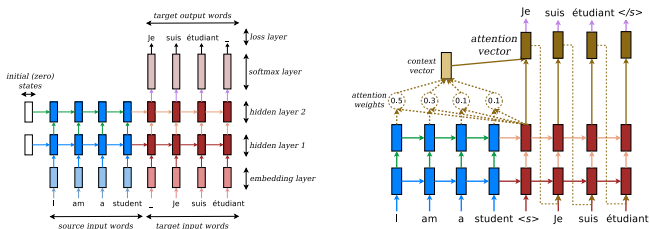


Figure: RNN (left) vs Attention (right) on a transformer (Chapter 15)⁹

⁹Kevin P. Murphy (2022). *Probabilistic Machine Learning: An introduction*.

How to make attention understand position

- Attention is permutation-invariant
- Absolute positional embedding (PE): Multiply input with functions

$$p_{i,2j} = \sin\left(\frac{i}{C^{2j/d}}\right), p_{i,2j+1} = \cos\left(\frac{i}{C^{2j/d}}\right)$$

- Rotational positional embedding (RoPE): Use rotation

$$\mathbf{R}_{\Theta,m}^d = \begin{pmatrix} \cos m\theta_1 & -\sin m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ \sin m\theta_1 & \cos m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cos m\theta_2 & -\sin m\theta_2 & \cdots & 0 & 0 \\ 0 & 0 & \sin m\theta_2 & \cos m\theta_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \cos m\theta_{d/2} & -\sin m\theta_{d/2} \\ 0 & 0 & 0 & 0 & \cdots & \sin m\theta_{d/2} & \cos m\theta_{d/2} \end{pmatrix}$$

Figure: RoPE Matrix¹⁰

¹⁰ Jianlin Su et al. (2023). *RoFormer: Enhanced Transformer with Rotary Position Embedding*. [arXiv: 2104.09864](https://arxiv.org/abs/2104.09864).

Extending Equivariance to Graphs

- Extending convolution to graphs via a graph Fourier transform
- For graph shift operator (usually adjacency or Laplacian) $\mathbf{S}$ with eigendecomposition $\mathbf{S} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$, define the filter as

$$\mathbf{x}^{(t+1)} := \rho(H_t(\mathbf{S}) \mathbf{x}^{(t)} + \mathbf{b}^{(t)} \mathbf{1}^\top), \quad H_t(\mathbf{S}) \mathbf{x} := \sum_{i=1}^N \mathbf{v}_i \mathbf{v}_i^\top \mathbf{x} H_t(\lambda_i)^\top$$

- Graph convolutional networks (GCN): Approximate the spectral functions with polynomials, giving

$$\mathbf{x}^{(t+1)} := \rho\left(\sum_{k=0}^K \mathbf{S}^k \mathbf{x}^{(t)} \mathbf{c}_k^\top + \mathbf{b}^{(t)} \mathbf{1}^\top\right).$$

Message-Passing Neural Networks

It can be shown (e.g. in Vasileiou et al. 2026) that GCNs are equivalent to message-passing GNNs, written as

$$\mathbf{x}_u^{(t+1)} := \text{UPD}_t\left(\mathbf{x}_u^{(t)}, \text{AGG}_t(\{\{\mathbf{x}_v^{(t)} \mid v \in N_G(u)\}\})\right)$$

- Each layer aggregates features from local neighborhoods
- After L layers, $\mathbf{x}_u^{(L)}$ depends only on the L -hop neighborhood of u .

Example on qLDPC codes

Tanner graphs are a common choice as the underlying graph for GNN layers.

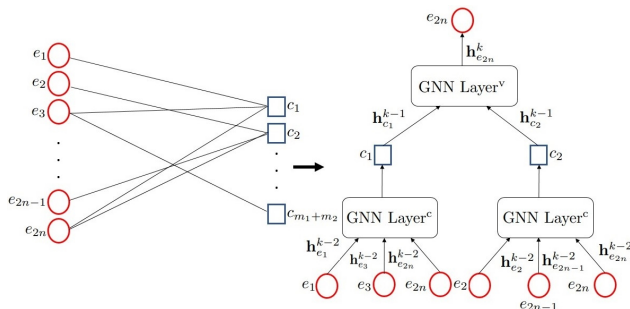


Figure: Example structure for decoding qLDPC codes (Fig. 1¹¹)

¹¹Vukan Ninkovic et al. (2024). *Decoding Quantum LDPC Codes Using Graph Neural Networks*. [arXiv: 2408.05170](https://arxiv.org/abs/2408.05170).

Example on qLDPC codes (continued)

The authors modeled belief propagation as a form of message-passing, motivating an MPNN.

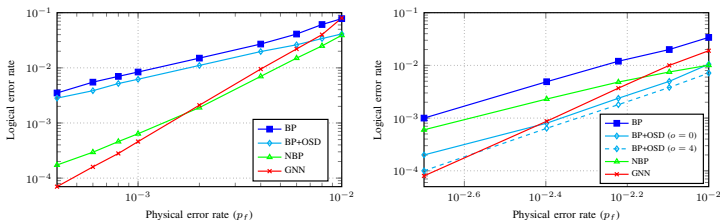


Figure: Decoding LER for example of Ninkovic et al. 2024 on hypergraph (left) and bicycle (right) codes (Figs. 2, 3¹²)

¹²Vukan Ninkovic et al. (2024). *Decoding Quantum LDPC Codes Using Graph Neural Networks*. [arXiv: 2408.05170](https://arxiv.org/abs/2408.05170).

Performance Metrics (review)

Metrics used to evaluate QEC Decoders in practice:

- Error threshold t
- Logical Error rate p_l . Note that after n rounds of correction,

$$P_n(E) = \frac{1}{2}(1 - (1 - 2p_l)^n)$$

- Decoding time (per cycle, and latency)

AlphaQubit 1: Overview

AlphaQubit 1¹³, developed by Google. Main ideas:

- Pretrain on synthetic data and fine-tune on scarce experimental data
- Takes factors into account such as leakage and analog stabilizer measurement outputs
- Architecture consists of recurrent network containing dilated CNNs and transformers

Results summary of AlphaQubit 1:

- LER at distance 3: $(2.901 \pm 0.023) \times 10^{-2}$.
- LER at distance 5: $(2.748 \pm 0.015) \times 10^{-2}$.
- Better than a tensor network (with 3.028 ± 0.023 LER at distance 3 and 2.915 ± 0.01 LER at distance 5)

¹³[Johannes Bausch et al. \(Nov. 2024\)](#). "Learning high-accuracy error decoding for quantum processors". In: *Nature*. DOI: [10.1038/s41586-024-08148-8](https://doi.org/10.1038/s41586-024-08148-8).

Overview (Training)

Circuit noise models:

- ❶ Circuit-related DEMs (up to 2 billion samples):
 - The (even) detection error event correlations p_{ij}
 - Device calibration data directly (XEB)
- ❷ SI1000 (500 million samples)

Operation	Noise Type	Strength
Z (X) Measurement	Preceded by bitflip (phaseflip) channel	$5p$
Z (X) Reset	Resets to $ 0\rangle$ ($ +\rangle$) and applies a bitflip (phaseflip) channel	$2p$
Resonator Idle	Applies 1q depolarizing channel to qubits not measured/reset	$2p$
2q Clifford	2q depolarizing channel	p
1q Clifford	1q depolarizing channel	$p/10$
Idle	1q depolarizing channel	$p/10$

Figure: Noise parameters for the SI1000 model (Extended data Fig. 1)

Finetuning: Partition 325,000 even experimental samples into training and validation sets. Odd samples held back as a test set.

Overview (continued)

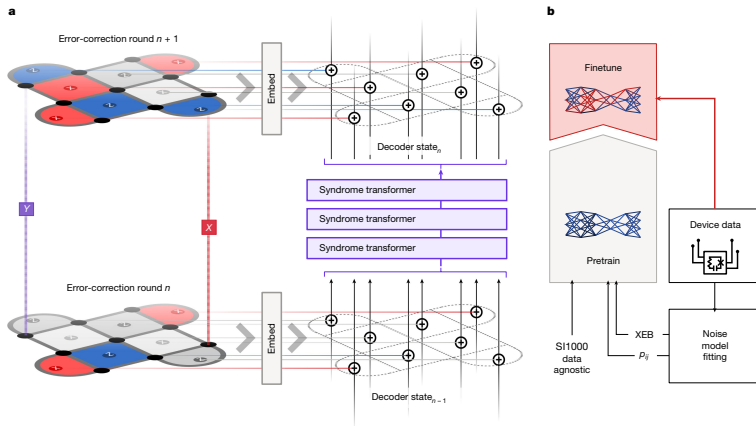


Figure: Overview of training method for AlphaQubit 1 (Fig. 2)

Architecture outline

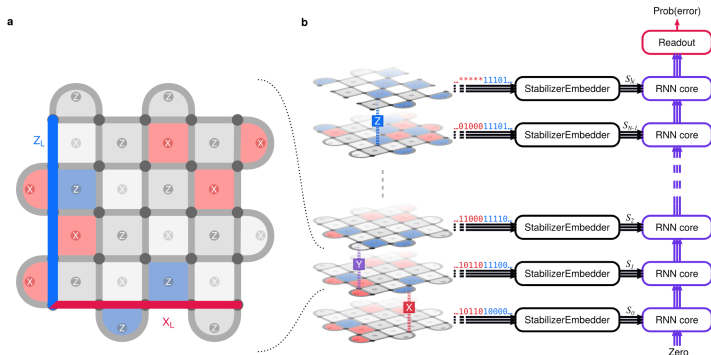


Figure: Architecture outline of AlphaQubit 1 (Extended Data Fig. 4)

Training details

- Increasing noise variance during training
- Fine-tuning (Sycamore data): Only 19 880 experimental samples used for training; 5 120 for early stopping
- Ensembling: Aggregating results from 20 identical models with different seeds to further improve LER

Performance

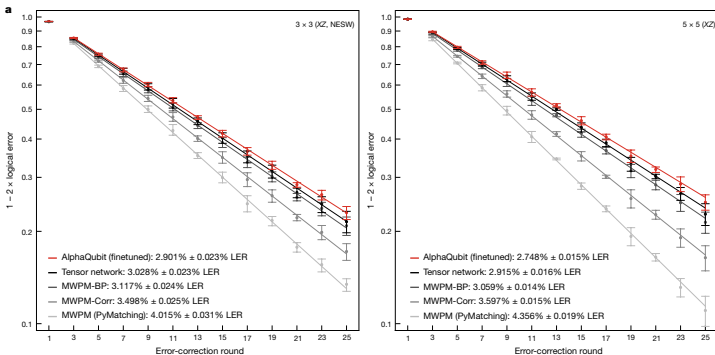


Figure: Logical error per round on 3×3 and 5×5 Sycamore experiment (Fig. 3)

Performance (continued)

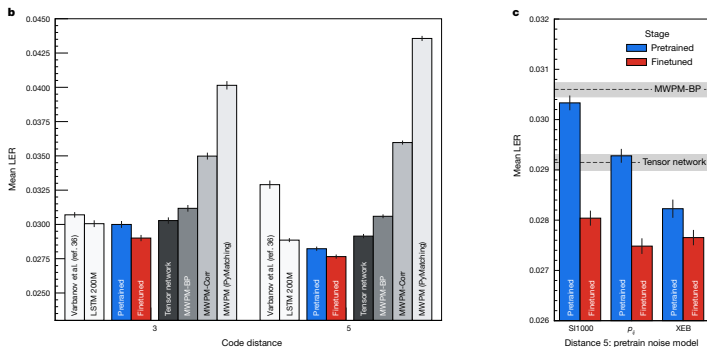


Figure: LERs of decoder, before and after fine-tuning on Sycamore experiments (Fig. 3)

Accuracy tradeoffs

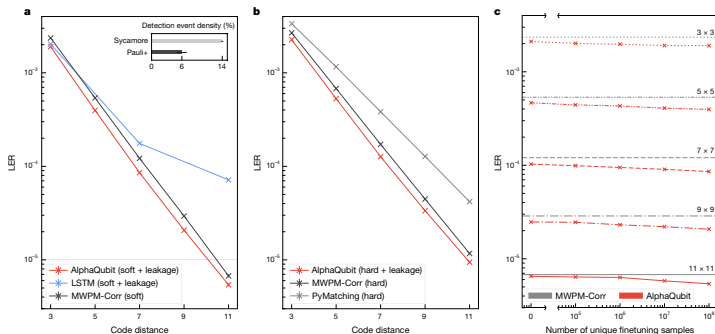


Figure: Accuracy tradeoffs: a,b) Effect of increasing code distance. Decoder is finetuned on 100 million samples from Pauli+ after pretraining on (SI1000). c) Accuracy vs. Pauli+ finetuning samples (Fig. 4)

Architecture details

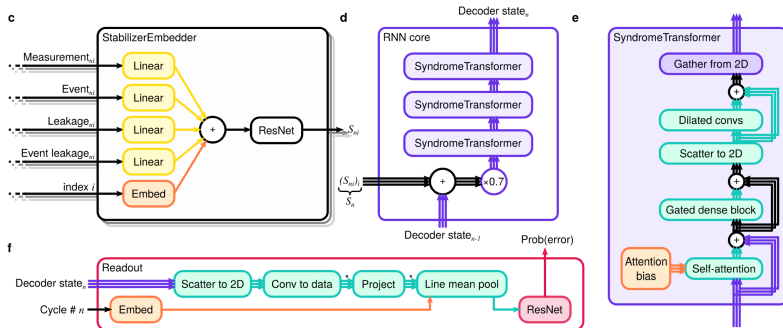


Figure: Details of Embedder and RNN core (Extended Data Fig. 4)

Ablation studies

The authors perform tests to measure effect of simplifying architecture. A selected list of simplifications:

- SimpleReadoutStack: Simplify the Readout ResNet.
- SimpleInputStack: Simplify embedding ResNet from 2 layers to 1 layer.
- FewerDims: Reduce the syndrome transformer embedding size
- FewerLayers: Reduce the syndrome transformer depth from 3 to 1
- OnlyEvents: Only provide detection events (remove measurements).
- OnlyMeasurements: Only provide stabilizer measurements (remove events).

Ablation studies (continued)

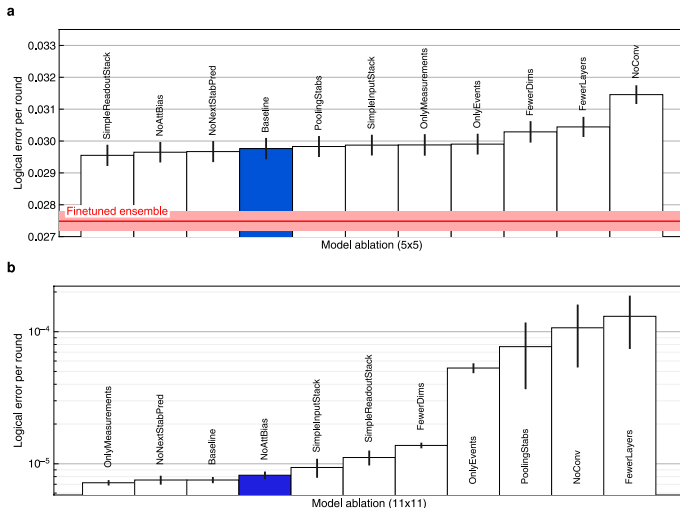


Figure: Effect of simplifying decoder structure: a) 5×5 DEM. b) 11×11

Pauli+

Ablation studies (continued)

c

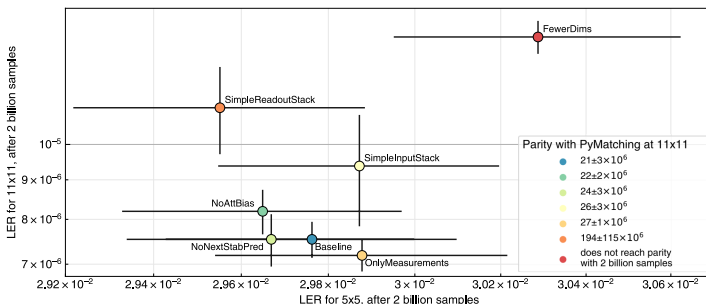


Figure: Effects of simplifying architecture for 11×11 compared with 5×5 surface code (Extended data Fig. 9)

Larger Error-Correction Rounds

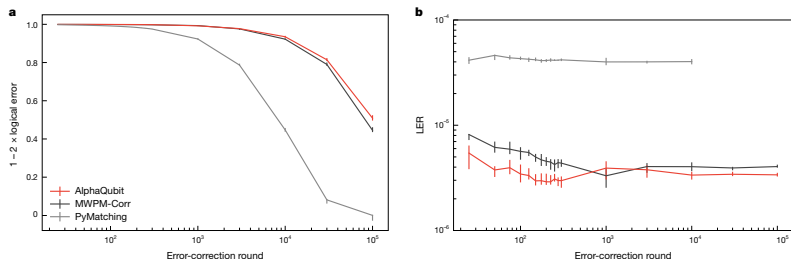


Figure: Comparison of AlphaQubit, MWPM-Corr and MWPM (PyMatching) on large number of error correction rounds. (Fig. 4)

Decoding time

- Authors admit they only optimized for reasonable training
- They propose a range of speedup techniques, once the feasibility has been proved.

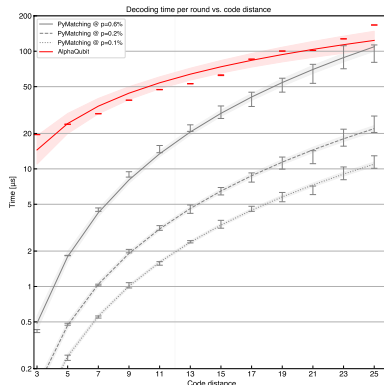


Figure: Decoding throughput vs code distance (Extended data Fig. 7)

AlphaQubit 2

AlphaQubit2¹⁴ trains a full version and a version for real-time decoding.

Main differences between AlphaQubit 1 and AlphaQubit 2-full are:

- Layer-based architecture with light-weight RNN layers instead of a large RNN core.
- Learned temporal compression of consecutive frames (3 to 6) before processing.
- No convolutions (replaced with spatial RoPE).
- Use of cross-attention at the start of the readout network.
- AQ2 was also applied to color-code decoding

¹⁴ [Andrew W. Senior et al. \(2026\)](#). *A scalable and real-time neural decoder for topological quantum codes*. [arXiv: 2512.07737](#).

AlphaQubit 2 Overview

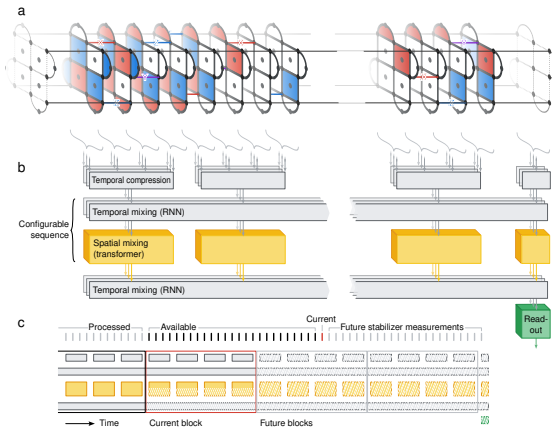


Figure: Overview of AlphaQubit 2 for surface-code decoding (Fig. 1)

AlphaQubit 2 Architecture detail

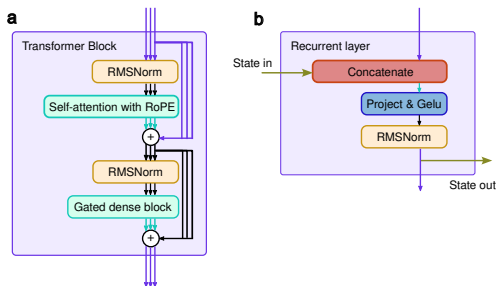


Figure: Building blocks in AlphaQubit 2 (Fig. S1)

Decoding Time Experiments

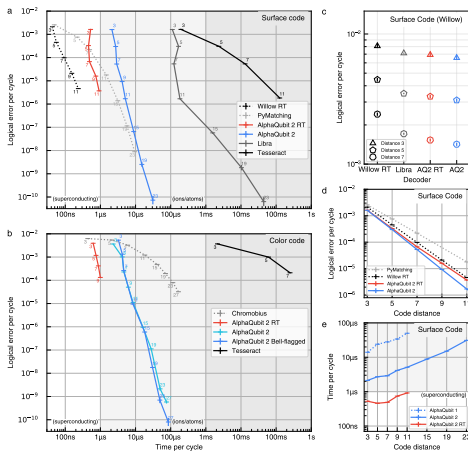


Figure: Decoding Throughput comparison (Fig. 4)

Decoding Time Experiments-Latency

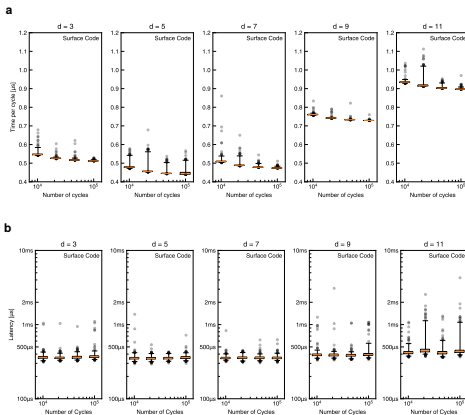


Figure: Latency and throughput experiments for AlphaQubit 2 (Fig. S11)

References I

-  Bausch, Johannes et al. (Nov. 2024). “Learning high-accuracy error decoding for quantum processors”. In: *Nature*. DOI: [10.1038/s41586-024-08148-8](https://doi.org/10.1038/s41586-024-08148-8).
-  Cao, Hanyan et al. (2026). *Maximum Likelihood Decoding of Quantum Error Correction Codes*. [arXiv: 2605.17230](https://arxiv.org/abs/2605.17230).
-  Cui, Ximin et al. (2019). “Multiscale Spatial-Spectral Convolutional Network with Image-Based Framework for Hyperspectral Imagery Classification”. In: *Remote Sensing* 11.19. DOI: [10.3390/rs11192220](https://doi.org/10.3390/rs11192220).
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